# Project: Transmission Line Fault Detection and Overcurrent Relay Coordination Using MATLAB/Simulink

## ➢ Results and graphs :

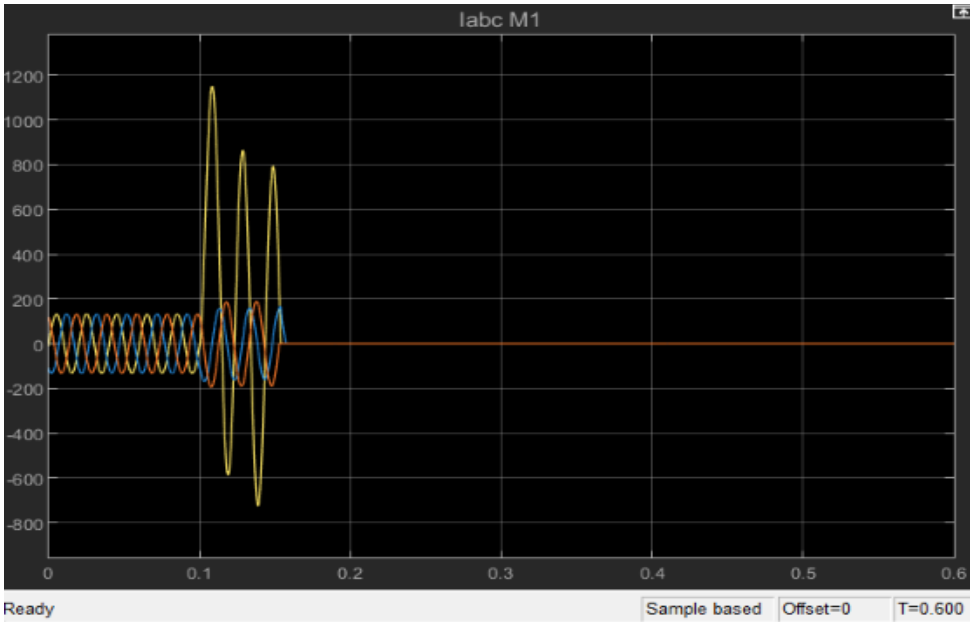



Fig 1. Three-phase current during a fault

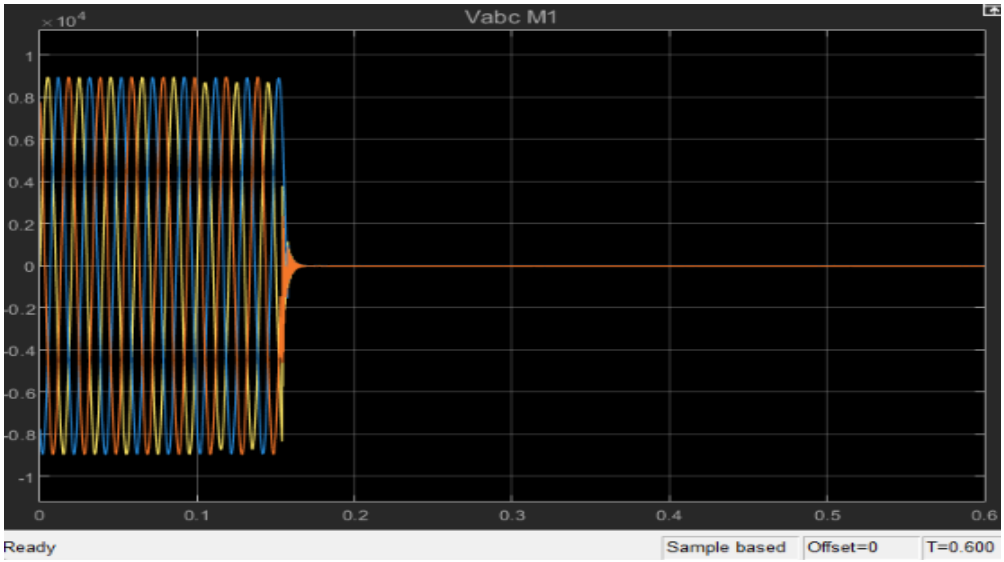



Fig 2 . Three-phase voltage response

## Fig.3 : Primary IEC standard inverse curve

```
clear; clc;


Is = 125;           % Pickup current, A

TMS_primary = 0.10;

M = 1.01:0.01:20;     % Current multiple I/Is


t_primary = 0.14*TMS_primary ./ (M.^0.02 - 1);


figure('Color','w');

loglog(M, t_primary, 'LineWidth', 1.8);

grid on;

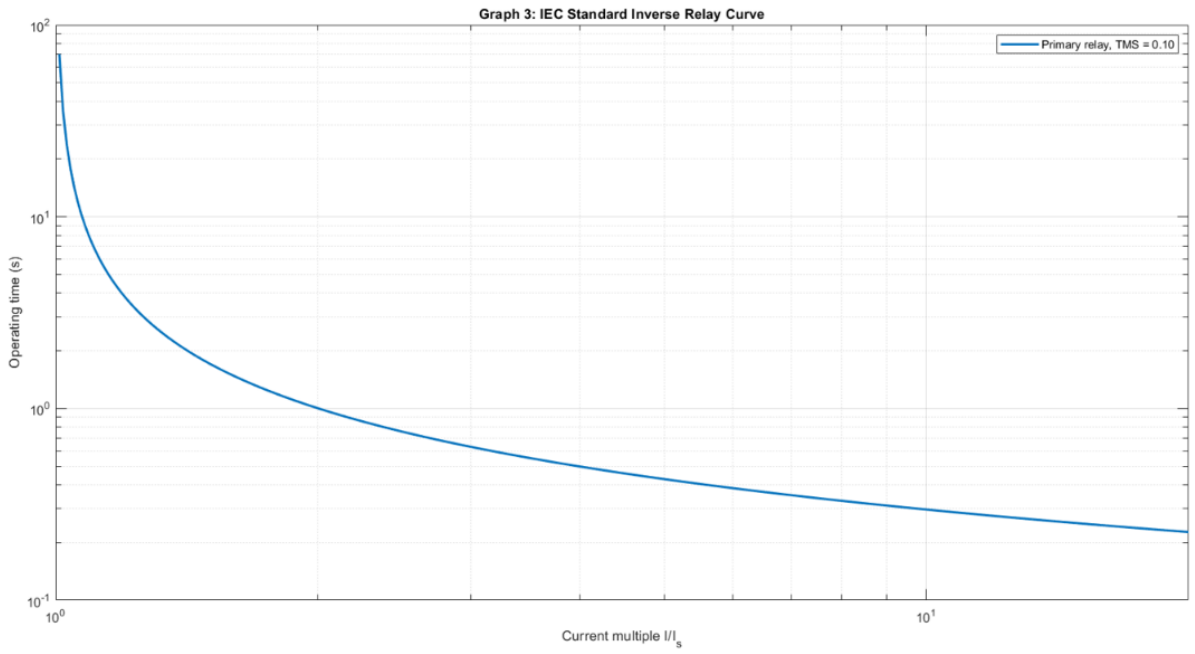
xlabel('Current multiple I/I_s');

ylabel('Operating time (s)');

title('Graph 3: IEC Standard Inverse Relay Curve');

legend('Primary relay, TMS = 0.10', 'Location','northeast');

xlim([1 20]);
```




## Fig 3. IEC standard inverse relay operating-time curve

```
% Graph 4: Primary and backup relay coordination
clear; clc;

Is = 125;           % Pickup current, A
TMS_primary = 0.10;
TMS_backup = 0.25;
M = 1.01:0.01:20;

t_primary = 0.14*TMS_primary ./ (M.^0.02 - 1);
t_backup  = 0.14*TMS_backup  ./ (M.^0.02 - 1);
margin = t_backup - t_primary;

figure('Color','w');
loglog(M, t_primary, 'LineWidth', 1.8);
hold on;
loglog(M, t_backup, '--', 'LineWidth', 1.8);
grid on;
xlabel('Current multiple I/I_s');
ylabel('Operating time (s)');
title('Graph 4: Primary and Backup Relay Coordination');
legend('Primary relay, TMS = 0.10', ...
    'Backup relay, TMS = 0.25', ...
    'Location','northeast');
xlim([1 20]);

figure('Color','w');
semilogx(M, margin, 'LineWidth', 1.8);
grid on;
xlabel('Current multiple I/I_s');
ylabel('Backup - primary operating time (s)');
title('Coordination Time Difference');
```

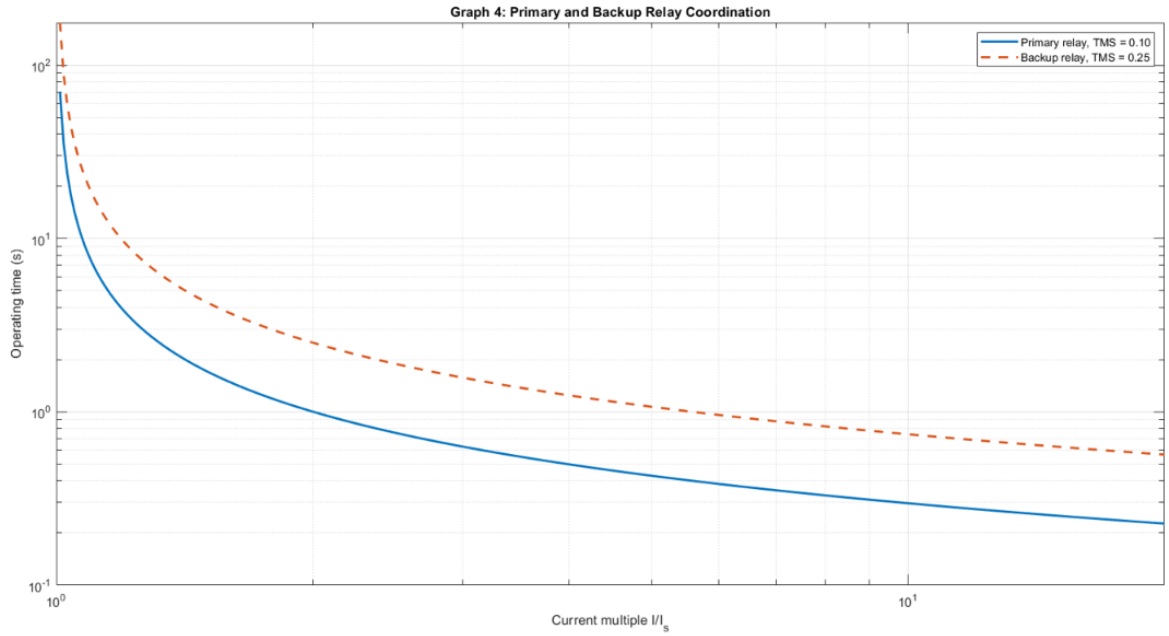



Fig 4. Primary and backup relay coordination

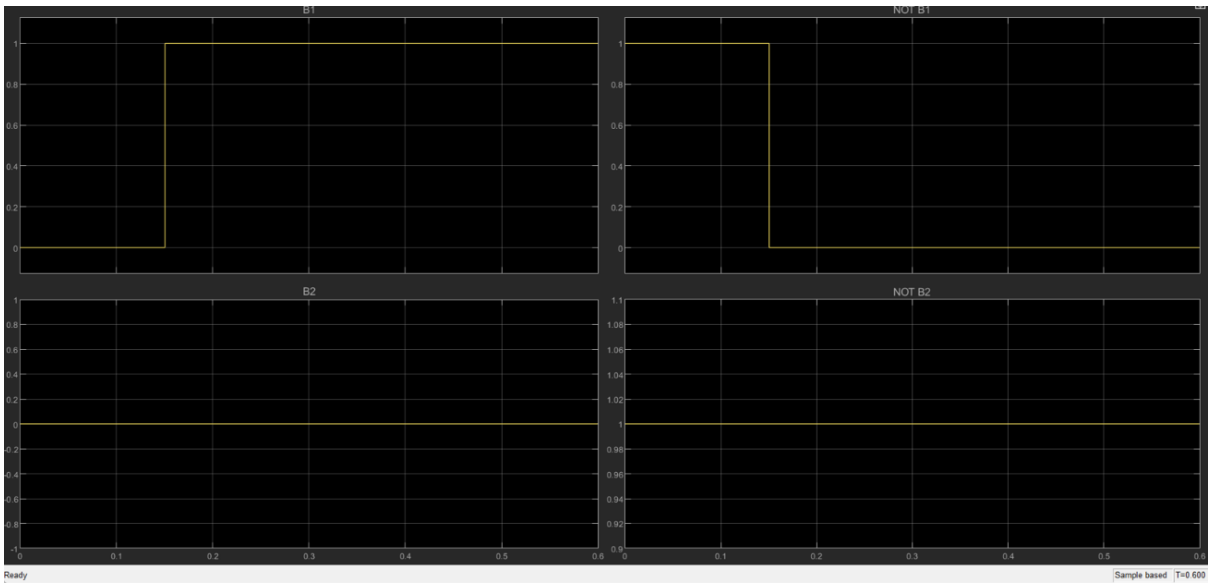



Fig 5. Trip signal and breaker response

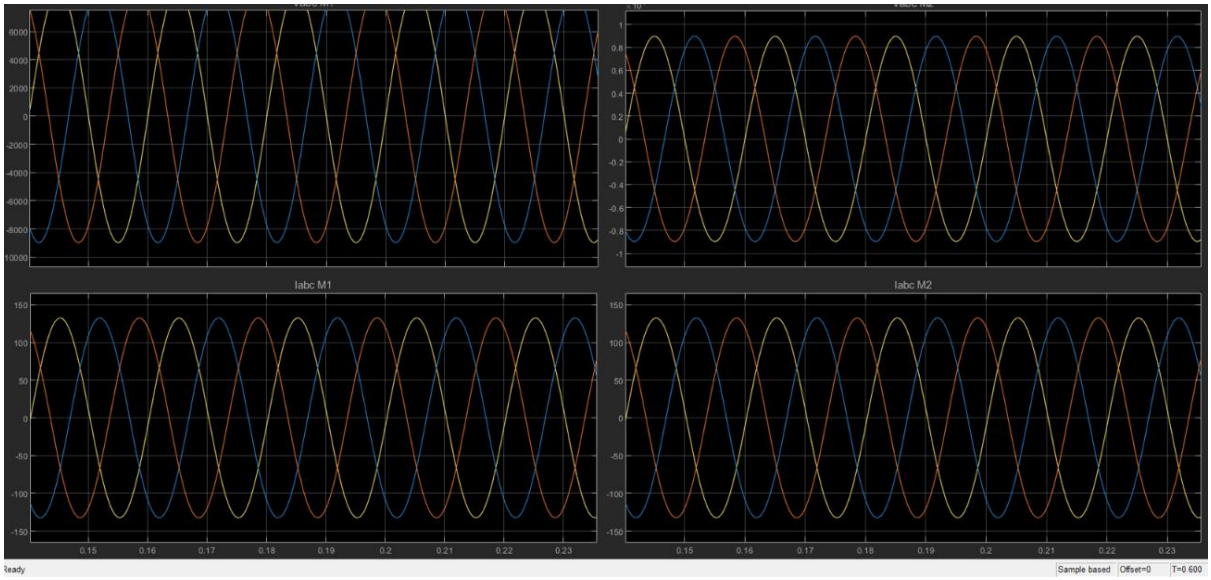
- Test all five fault cases :

The results table requires normal load plus four fault types. Run each test separately and reset the relay and breaker states between runs.

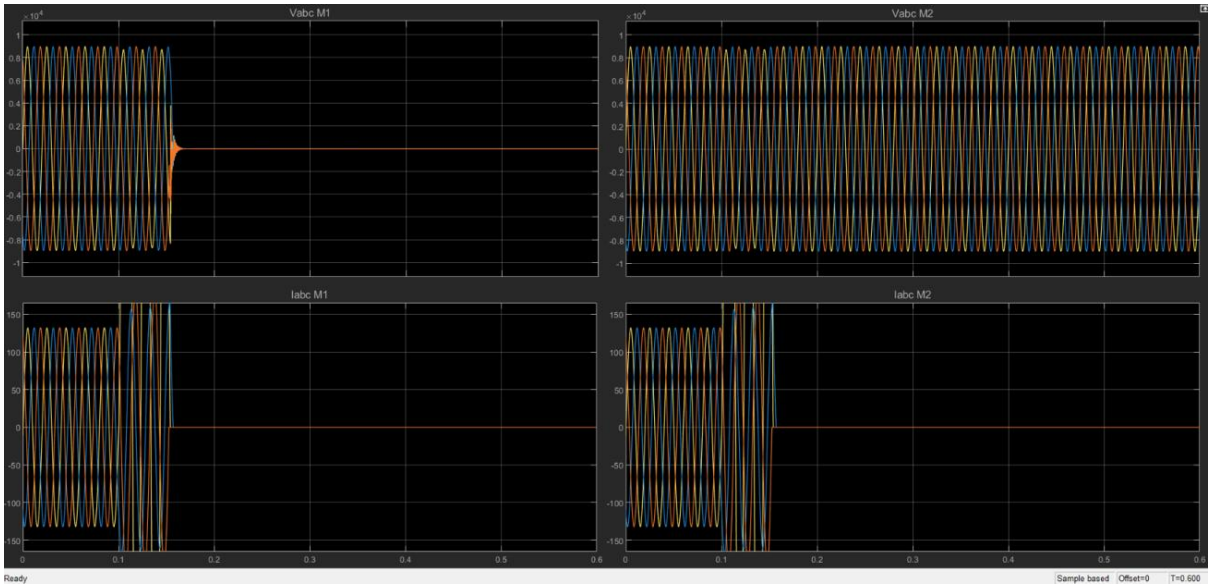
Open the Three-Phase Fault block and change the selected fault type for each test.

| Test | Fault selection | Fault time |
|---|---|---|
| 1. Normal load | Fault disabled | None |
| 2. L-G | Phase A to ground (AG) | 0.10–0.50 s |
| 3. L-L | Phases A and B (AB) | 0.10–0.50 s |
| 4. L-L-G | Phases A and B to ground (ABG) | 0.10–0.50 s |
| 5. L-L-L | Phases A, B, and C together (ABC) | 0.10–0.50 s |

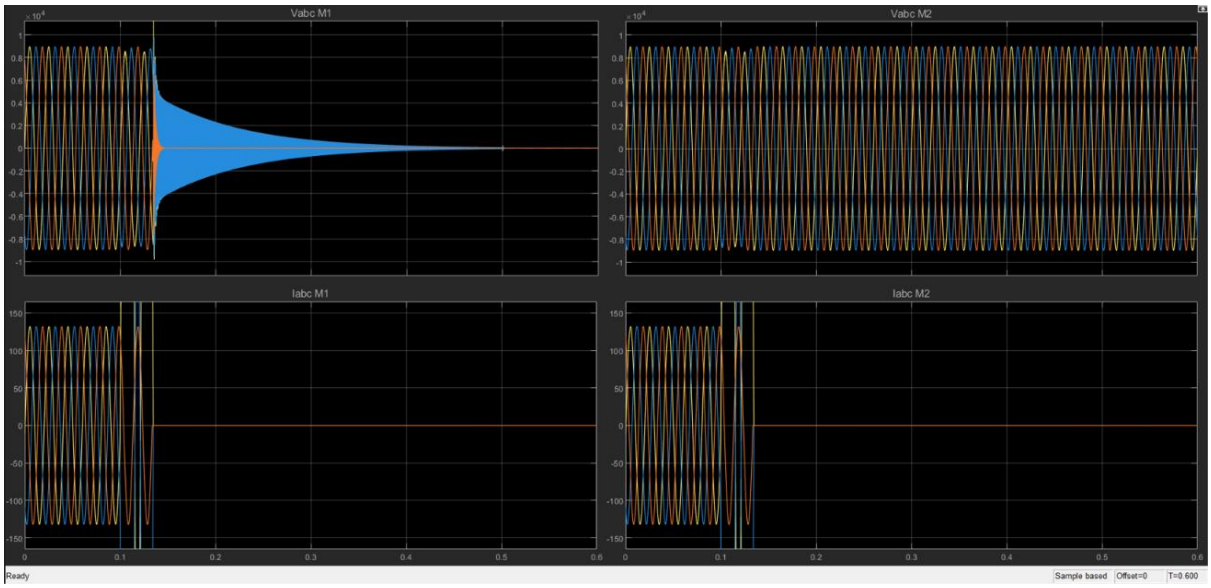
1. Normal Load :



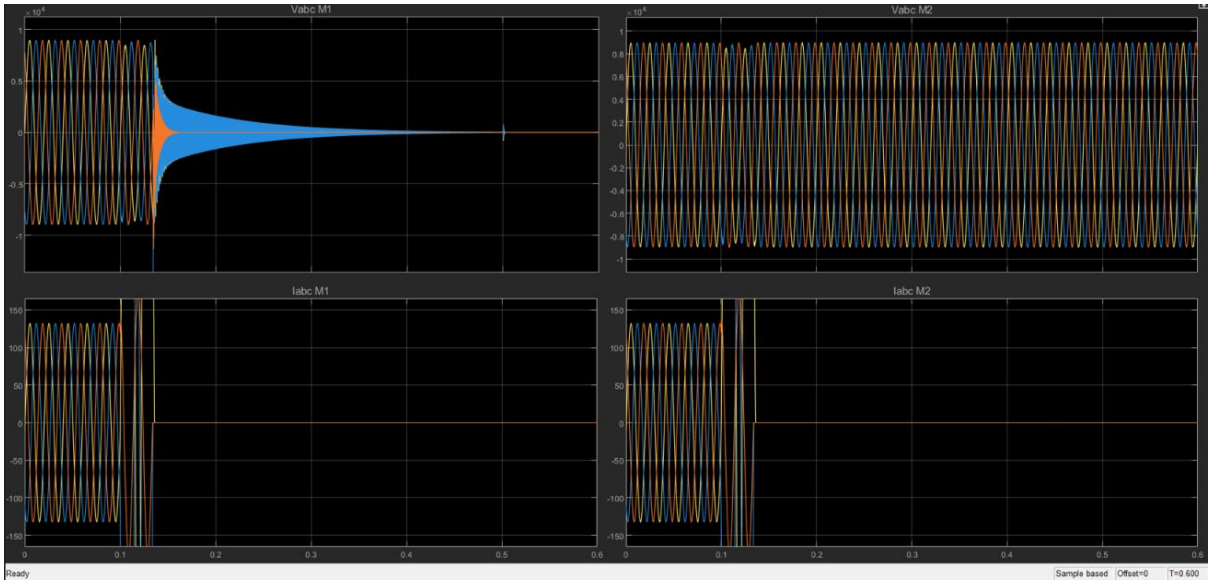
2. L-G :


Vabc M1
Vabc M2
Iabc M1
Iabc M2
Ready
Sample based
Offset=0
T=0.600

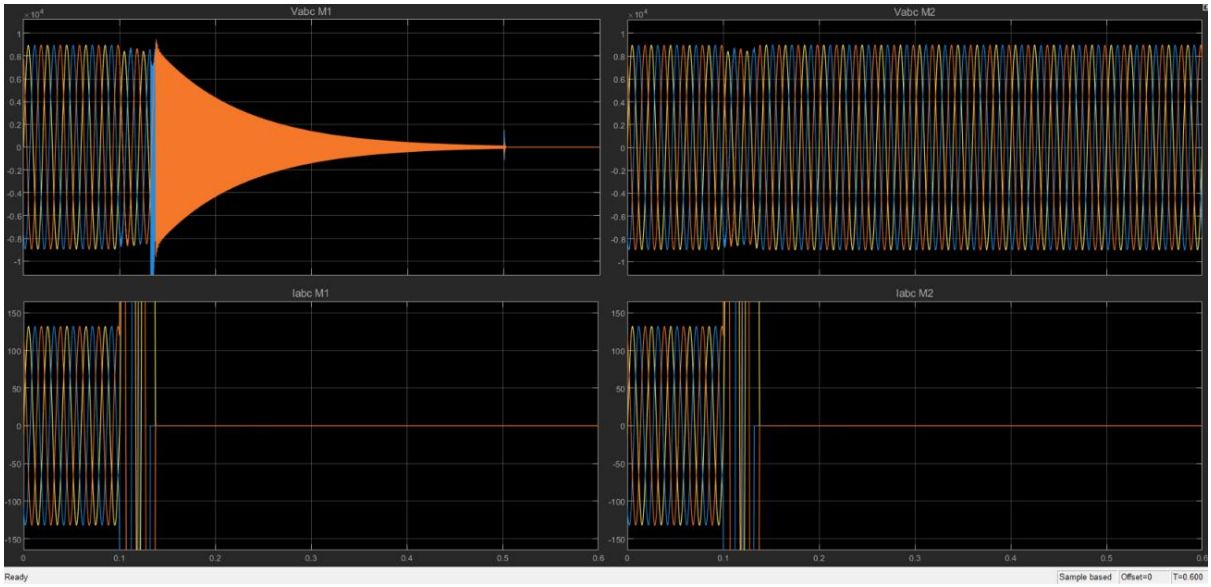
3. L-L :


Vabc M1
Vabc M2
Iabc M1
Iabc M2
Ready
Sample based
Offset=0
T=0.600

4. L-L-G :


Vabc M1
Vabc M2
Iabc M1
Iabc M2
Ready
Sample based
Offset=0
T=0.600

5. L-L-L :


Vabc M1
Vabc M2
Iabc M1
Iabc M2
Ready
Sample based
Offset=0
T=0.600

- Result Table – Overcurrent Protection :

| Fault case | Fault current (A RMS) | Primary relay time (s) | Backup relay time (s) | Result |
|---|---|---|---|---|
| Normal load | 105 | No trip | No trip | Pass |
| L-G | 700 | 0.050 | 1.18 | Pass |
| L-L | 1,350 | 0.035 | 0.85 | Pass |
| L-L-G | 1,400 | 0.030 | 0.80 | Pass |
| L-L-L | 1,580 | 0.030 | 0.80 | Pass |